



# MAULANA ABUL KALAM AZAD UNIVERSITY OF TECHNOLOGY, WEST BENGAL

Paper Code : PEC-CS702E/PEC-IT702F Cyber Security

UPID : 007558

Time Allotted : 3 Hours

Full Marks : 70

The Figures in the margin indicate full marks.

Candidate are required to give their answers in their own words as far as practicable

## Group-A (Very Short Answer Type Question)

1. Answer *any ten* of the following :

[ 1 x 10 = 10 ]

- (I) Which method of hacking will record all your keystrokes?
- (II) Which malware does not clone or replicate itself through infection?
- (III) Which of the following actions compromise cyber security?  
a) Vulnerability    b) Attack    c) Threat    d) Exploit
- (IV) Write the utility of Passwords.
- (V) "Cyberspace" was coined by \_\_\_\_\_.
- (VI) Which of these is NOT a scanning type?  
a. Null Scan    b. Xmas Tree Scan    c. SYN Stealth    d. Cloud Scan
- (VII) Which type of attack is an internet scam done by cyber-criminals where the user is convinced digitally to provide confidential information?
- (VIII) Which port number is usually referred to as the port number (default) of several web servers like apache?
- (IX) Name a hacking approach where cyber-criminals design fake websites or pages for tricking or gaining additional traffic.
- (X) Which of the following is not a step followed by cyber-criminals in data breaching?  
a) Exfiltration    b) Research and info-gathering    c) Attack the system    d) Fixing the bugs
- (XI) Name the world's very first antivirus program.
- (XII) IT security in any firm or organization is maintained and handled by \_\_\_\_\_.

## Group-B (Short Answer Type Question)

Answer *any three* of the following :

[ 5 x 3 = 15 ]

2. Explain data leakage. [5]
3. (i) Define Hacker. List Out the Types of hackers [5]  
(ii) What is a Backdoor?
4. What is the difference between vulnerability assessment (VA) and penetration testing (PT)? [5]
5. Describe briefly the CIA Triad and how to achieve it. [5]
6. Explain the difference between hashing and encryption. [5]

## Group-C (Long Answer Type Question)

Answer *any three* of the following :

[ 15 x 3 = 45 ]

7. (i) Explain briefly about the Forensics Investigation Process. [ 8+7 ]  
(ii) Discuss the challenges in computer forensics.
8. What do you mean by Information System? What is the need of Information System? What are the types of Information System? [ 4+ 4+ 7 ]
9. Write a short note on [ 5+5+5 ]  
a) Patent Law    b) Copy write Law    c) IPR
10. (i) Discuss cybercrime and the Indian ITA 2000. [ 7+8 ]  
(ii) Explain in detail about architecture of IP Security
11. With a neat diagrams, Explain PGP Cryptographic Functions or PGP Functions (Authentication, Confidentiality, Confidentiality and Authentication). [ 15 ]

\*\*\* END OF PAPER \*\*\*